Part of Accenture

SCHOOL OF ARTIFICIAL INTELLIGENCE

AI Programming with Python

Syllabus

UDACITY.COM

Overview

Develop a strong foundation in Python programming for AI, utilizing tools like NumPy, pandas, and Matplotlib for data analysis and visualization. Learn how to use, build, and train machine learning models with popular Python libraries. Implement neural networks using PyTorch. Gain practical experience with deep learning frameworks by applying your skills through hands-on projects. Explore generative AI with Transformer neural networks, learn to build, train, and deploy them with PyTorch, and leverage pre-trained models for natural language processing tasks. Designed for individuals with basic programming experience, this program prepares you for advanced studies in AI and machine learning, equipping you with the skills to begin a career in AI programming.

Nanodegree Program | Beginner | 52 hours | 4.7 (633 Reviews)

Prerequisites

Prior to enrolling, you should have the following knowledge:

Elementary algebra | Linear algebra | Basic github | Basic calculus

You will also need to be able to communicate fluently and professionally **in written and spoken English**.

Skills You'll Learn

Beginner Python PCEP™ Certification Prep

Python functions | Python methods | Text processing in Python | Functional Python | Boolean expressions | Python operators | Basic Python | List comprehension | Python syntax | Python data types | Jupyter notebooks | Python best practices | Python variables | Control flow in Python | Python Certified Entry-Level Programmer | Python string methods | Python exception handling | Built-in Python functions | Python function definition | Python data structures | Python collections

Numpy, Pandas, Matplotlib

Python package management | Pandas | Pip | Jupyter notebooks | Anaconda | matplotlib | NumPy | Python packaging

Neural Networks - AI Programming with Python

Logistic regression | Deep learning framework proficiency | Classification models | Feedforward neural networks | Deep learning | Transfer learning | Training neural networks | Neural network basics | Basic PyTorch | Gradient descent | Perceptron | Neural network mechanics | Backpropagation | PyTorch

Programming Transformers with PyTorch

Generative AI Awareness | Text generation | Attention mechanisms | GPT | Hugging Face | Transformer neural networks | Foundation Model Concepts | Word embeddings | PyTorch | Natural language processing | NLP transformers

Courses

01. Introduction to AI Programming

2 hours

Welcome to the AI programming with python Nanodegree Program!
Come and explore the beautiful world of AI.

L1	Welcome to AI Programming with Python	Welcome to the AI Programming with Python Nanodegree program!
L2	Getting Help	You are starting a challenging but rewarding journey! Take 5 minutes to read how to get help with projects and content.
L3	Get Help with Your Account	What to do if you have questions about your account or general questions about the program.

02. Beginner Python PCEP™ Certification Prep

21 hours

Prepare for the Python PCEP certification while deepening your skills in syntax, functions, control flow, and more. This course builds a strong foundation for coding and career readiness.

L1	Python Programming Fundamentals	Start your journey into the world of Python programming. Define fundamental terms and understand Python's logical structure. Learn how to declare and use variables to store different types of data.
L2	Python Boolean Logic and Numeric Operators	Learn the fundamentals of Boolean logic and numeric operations in Python. Explore Boolean operators, relational operators, type casting, I/O operations, and limitations of floating-point numbers.
L3	Python Control Flow and Loops	Master the art of decision-making and repetition in Python. Explore conditional statements like if, elif, and else to create branching logic in your programs.
L4	Python Data Structures	Explore fundamental Python data structures like lists, tuples, and dictionaries. Learn to create, manipulate, and iterate over lists, understanding indexing, slicing, and list comprehensions.
L5	Python Functions	Explore functions and function handling. You'll learn to break complex problems into smaller, reusable functions.

L6	Python Exceptions	Explore Python's exception handling mechanism, common exception types, and how to handle them using try-except blocks. Learn advanced techniques like propagating and delegating exceptions.
L7	Project: Project: A Data-Driven Approach to AI and Python	Solidify your understanding of data manipulation, exploration, and presentation as you explore the role of a data scientist working with Amazon Reviews to transform raw data into valuable insights.

03. Numpy, Pandas, Matplotlib

8 hours

This course covers library packages for Python, such as Numpy (which adds support for large data), Pandas (which is used for data manipulation and analysis) and Matplotlib (which is used for data visualization).

L1	Anaconda	Anaconda is a package and environment manager built specifically for data. Learn how to use Anaconda to improve your data analysis workflow.
L2	Jupyter Notebooks	Jupyter Notebooks are a great tool for getting started with writing python code. Though in production you often will write code in scripts, notebooks are wonderful for sharing insights and data viz!
L3	NumPy	Learn the basics of NumPy and how to use it to create and manipulate arrays.
L4	Pandas	Learn the basics of Pandas Series and DataFrames and how to use them to load and process data.
L5	Matplotlib and Seaborn Part 1	Learn how to use matplotlib and seaborn to visualize your data. In this lesson, you will learn how to create visualizations to depict the distributions of single variables.
L6	Matplotlib and Seaborn Part 2	In this lesson, you will use matplotlib and seaborn to create visualizations to depict the relationships between two variables.

04. Neural Networks - AI Programming with Python

13 hours

This course on neural networks explains how algorithms inspired by the human brain operate and puts to use those concepts when designing neural networks to solve particular problems.

L1	Welcome to Neural Networks	Get started with neural networks: foundations, perceptrons, deep learning, PyTorch, and practical skills. Meet your instructors and set up your learning environment.

L2	Introduction to Neural Networks	In this lesson, Luis will give you solid foundations on deep learning and neural networks. You'll also implement gradient descent and backpropagation in Python right here in the classroom.
L3	Implementing Gradient Descent	Mat will introduce you to a different error function and guide you through implementing gradient descent using numpy matrix multiplication.
L4	Training Neural Networks	Now that you know what neural networks are, in this lesson you will learn several techniques to improve their training.
L5	Deep Learning with PyTorch	Learn how to use PyTorch for building deep learning models.

05. Programming Transformers with PyTorch

9 hours

This course will guide you through the essential concepts of Transformer Neural Networks and their implementation using PyTorch. Starting with an introduction to Transformers, you will learn to build and train Transformer models from scratch. Additionally, you will explore the advantages of using pre-trained Transformer models and how to leverage them effectively in your projects. By the end of this course, you will have a solid foundation in programming Transformer Neural Networks with PyTorch.

L1	Introduction to Transformer Neural Networks	Explore Transformer neural networks, their architecture, and applications like ChatGPT. Delve into NLP basics, tokenization, and model training using PyTorch for AI advancements.
L2	Building Transformer Neural Networks with PyTorch	Learn to build a Transformer model with PyTorch, covering tokenization, embeddings, multi-head attention, training, and text generation for NLP tasks.
L3	Using Pre-Trained Transformers	Master the use of pre-trained transformers, including their training, fine-tuning, limitations, and applying them to NLP tasks like text generation and QA.
L4	Project: Project: Transformers for Movie Review Sentiment Analysis	Develop a transformer-based model for movie review sentiment analysis, achieving over 75% accuracy through data preparation, custom dataset implementation, and effective training loops.

Meet Your Instructors

Brian Hough

AWS DevTools Hero & Founder of Tech Stack Playbook®

Brian H. Hough is an AWS DevTools Hero and founder of Tech Stack Playbook®, where he builds tech for clients and creates educational content for 20k+ followers. His career began at the Harvard Kennedy School, then in VC, won 5 global hackathons & has worked with AWS, freeCodeCamp, The Motley Fool, Boston University & more.

Mat Leonard

Content Developer

Mat is a former physicist, research neuroscientist, and data scientist. He did his PhD and Postdoctoral Fellowship at the University of California, Berkeley.

Juan Delgado

Computational Physicist

Juan is a computational physicist with a Masters in Astronomy. He is finishing his PhD in Biophysics. He previously worked at NASA developing space instruments and writing software to analyze large amounts of scientific data using machine learning techniques.

Juno Lee

Data Scientist

As a data scientist at Looplist, Juno built neural networks to analyze and categorize product images, a recommendation system to personalize shopping experiences for each user, and tools to generate insight into user behavior.

Mike Yi

Mathematician

Mike is a mathematician with a PhD in Cognitive Science from the University of California, Irvine.

Luis Serrano

ML Engineer

Luis was formerly a Machine Learning Engineer at Google. He holds a PhD in mathematics from the University of Michigan, and a Postdoctoral Fellowship at the University of Quebec at Montreal.

Ivan Mushketyk

Software Engineer

Experienced software engineer with over ten years in the field. Worked at at AWS and Stripe, as well as several start-ups. An experienced online instructor, creating courses on AI, data engineering, and AWS. Passionate about leveraging technology to solve complex problems and sharing knowledge with others.

Why Udacity

Demonstrate proficiency with practical projects

Projects are based on real-world scenarios and challenges, allowing you to apply the skills you learn to practical situations, while giving you real hands-on experience

- Gain proven experience
- Retain knowledge longer
- Apply new skills immediately

24/7 access to real human support

Reviewers provide timely and constructive feedback on your project submissions, highlighting areas of improvement and offering practical tips to enhance your work

- Get help from subject matter experts
- Gain valuable insights and improve your skills
- Learn industry best practices